EE-3TR4 Mid Term Exam 2022

Duration of examination: 1 hour and 30 minutes
Answer all the questions.
Total marks = 20.

1. Find the Fourier transform of the following signal
$$x(t) = 4\ \mathrm{sinc}^2[2(t-5)]\sin[6\pi(t-5)],$$
and sketch the magnitude spectrum, $|X(f)|$.
Hint: Use Fourier transform pairs and properties provided on the formula sheet.
7 marks
2. A 50% duty cycle square wave of amplitude 1 V and period 0.001 s, passes through an ideal low pass filter with the following transfer function,
$$H(f) = \mathrm{rect}\left(\frac{f}{2f_c}\right),$$
where the cutoff frequency, f_c = 2 kHz. Find
   a. The input magnitude spectrum (i.e. $|C_n|$ versus nf_0) and sketch it. Show up to third harmonics. 3 *marks*
   b. The output magnitude spectrum and sketch it. 2 *marks*
   c. The filter output in time domain. 2 *marks*

   Note: For (a), you need to calculate C_n using the formula shown in the formula sheet. You may also work with Fourier transform instead of Fourier series. In that case, replace C_n with the Fourier transform.
7 marks
3. Find the ratio of the lower sideband power to the carrier power in a standard AM with single-tone modulation. The message signal is of the form:
$$m(t) = A_m \sin(2\pi f_m t).$$
Assume that the modulation is 50%, A_c = 1 V, and the carrier frequency, $f_c \gg f_m$.
Sketch the modulated signal in time domain. Find the Fourier transform of the modulated wave and plot the magnitude spectrum.

   6 *marks*

   Mark distribution: Ratio calculation – 2 *marks*
   Time domain sketching – 1.5 *marks*
   Frequency domain calculation and sketching – 2.5 *marks*

Useful Information

Q1. Find the Fourier transform of the following signal

$$x(t) = 4\,\mathrm{sinc}^2[2(t-5)]\sin[6\pi(t-5)],$$

and sketch the magnitude spectrum, $|X(f)|$.

Hint: Use Fourier transform pairs and properties provided on the formula sheet.

$$x(t) = \underbrace{4\,\mathrm{sinc}^2[2(t-5)]}\,\underbrace{\sin[6\pi(t-5)]}$$
$$= x_1(t)\cdot x_2(t)$$

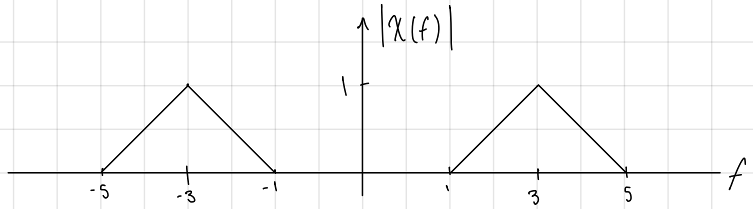
$$\mathrm{sinc}^2(t) \longleftrightarrow 2\,\mathrm{triang}\left(\tfrac{f}{2}\right)$$
$$4\,\mathrm{sinc}^2(t) \longleftrightarrow 2\cdot 2\,\mathrm{triang}\left(\tfrac{f}{2}\right)$$
$$x_1(t) = 4\,\mathrm{sinc}^2(2t) \longleftrightarrow \tfrac{1}{2}(2)(2)\,\mathrm{triang}\left(\tfrac{1}{2}\cdot\tfrac{1}{2}\cdot f\right) = 2\,\mathrm{triang}\left(\tfrac{1}{4}f\right) = X_1(f)$$

$$x_1(t) = \cancel{4\,\mathrm{sinc}^2[2(t-5)]} \longleftrightarrow \cancel{\tfrac{1}{2}(4)(2)\,\mathrm{triang}\left(\tfrac{1}{2}\cdot\tfrac{1}{2}\cdot f\right)\cdot e^{-j2\pi 5f}}$$
$$= \cancel{4\,\mathrm{triang}\left(\tfrac{1}{4}f\right)\cdot e^{-j10\pi f}}$$
$$= \cancel{X_1(f-f_c)\ ,\ f_c = 5}$$

$\tau = 2$,

$$\tau\,\mathrm{sinc}^2(\tau x) \longleftrightarrow \mathrm{triang}\left(\tfrac{x}{\tau}\right)$$
$$\tau\,\mathrm{sinc}^2(f\tau) \longleftrightarrow \mathrm{triang}\left(\tfrac{t}{\tau}\right)$$
$$2\,\mathrm{sinc}^2(2t) \longleftrightarrow \mathrm{triang}\left(\tfrac{t}{2}\right)$$
$$4\,\mathrm{sinc}^2(2t) \longleftrightarrow 2\,\mathrm{triang}\left(\tfrac{t}{2}\right)$$

$$= x_1(t)\cdot x_2(t)$$
$$= x_1(t)\cdot \sin[6\pi t]$$
$$= x_1(t)\cdot \sin[2\pi(3)t]\ ,\quad f_c = 3$$
$$X(f) = \tfrac{1}{2j}\left[X_1(f-f_c) + X_1(f+f_c)\right]$$
$$= \tfrac{1}{j}\cdot 2\left[\mathrm{triang}\left(\tfrac{1}{4}(f-3)\right) - \mathrm{triang}\left(\tfrac{1}{4}(f+3)\right)\right]$$
$$= \tfrac{1}{j}\left[\mathrm{triang}\left(\tfrac{1}{4}(f-3)\right) - \mathrm{triang}\left(\tfrac{1}{4}(f+3)\right)\right]\cdot e^{-j2\pi 5f} \longleftarrow \text{time shift property}$$
$$= \tfrac{1}{j}\left[\mathrm{triang}\left(\tfrac{1}{4}(f-3)\right) - \mathrm{triang}\left(\tfrac{1}{4}(f+3)\right)\right]\cdot e^{-j10\pi f}$$
$$|X(f)| = \left|\cancel{\tfrac{1}{j}}^{\,1}\left[\mathrm{triang}\left(\tfrac{1}{4}(f-3)\right) + \mathrm{triang}\left(\tfrac{1}{4}(f+3)\right)\right]\cdot \cancel{e^{-j10\pi f}}^{\,1}\right|$$
$$= \mathrm{triang}\left(\tfrac{1}{4}(f-3)\right) + \mathrm{triang}\left(\tfrac{1}{4}(f+3)\right)$$



Fourier Series

If $x(t)$ is a periodic signal with period T_0, it may be written as

$$x(t) = \sum_{n=-\infty}^{\infty} C_n \exp(j2\pi n f_0 t),$$

$$C_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) \exp(-j2\pi n f_0 t) dt,$$

and $f_0 = 1/T_0$.

Fourier Transforms

$\mathrm{rect}(t) \rightleftharpoons \mathrm{sinc}(f)$

$\mathrm{triang}(t) \rightleftharpoons \mathrm{sinc}^2(f)$

$\delta(t) \rightleftharpoons 1$

If $g(t) \rightleftharpoons G(f)$ and $h(t) \rightleftharpoons H(f)$, then

Scaling Property: $g(at) \rightleftharpoons \dfrac{1}{|a|} G(f/a)$

Duality Property: $G(t) \rightleftharpoons g(-f)$

Time Shifting Property: $g(t-t_0) \rightleftharpoons G(f) \exp(-j2\pi f t_0)$

Freq. Shifting Property: $g(t) \exp(j2\pi f_c t) \rightleftharpoons G(f - f_c)$

Corollary 1 of Freq. Shifting Property: $g(t) \cos(2\pi f_c t) \rightleftharpoons \dfrac{1}{2}[G(f-f_c) + G(f+f_c)]$

Corollary 2 of Freq. Shifting Property: $g(t) \sin(2\pi f_c t) \rightleftharpoons \dfrac{1}{2j}[G(f-f_c) - G(f+f_c)]$

Convolution theorem:

$g(t) * h(t) \rightleftharpoons G(f)H(f)$

$g(t)h(t) \rightleftharpoons G(f) * H(f)$

Filters :

$x(t) * h(t) = y(t)$

$X(f)H(f) = Y(f)$

AM :

standard AM: $s(t) = A_c[1 + k_a m(t)] \cos(2\pi f_c t)$

DSB-SC: $s(t) = A_c m(t) \cos(2\pi f_c t)$

Trigonometric Identities :

$2\cos(A)\cos(B) = \cos(A-B) + \cos(A+B)$

Set A=B, to find $\cos^2(A)$.

$2\sin A \sin B = \cos(A-B) - \cos(A+B)$

$2\sin A \cos B = \sin(A-B) + \sin(A+B)$

Mean Power of a sinusoid :

If a signal is of the $A\sin(2\pi f t + \theta)$, the mean power of the signal is $A^2/2$. The mean power does not depend on θ or f.

$\tau\, \mathrm{sinc}^2(\tau x) \longleftarrow \mathrm{triang}\left(\frac{x}{\tau}\right)$
$\tau\, \mathrm{sinc}^2(f\tau) \longleftarrow \mathrm{triang}\left(\frac{t}{\tau}\right)$
$\tau=2$, $2\, \mathrm{sinc}^2(2t) \longleftarrow \mathrm{triang}\left(\frac{t}{2}\right)$
$4\, \mathrm{sinc}^2(2t) \longleftarrow 2\, \mathrm{triang}\left(\frac{t}{2}\right)$

MIDTERM SOLUTIONS

1.

CONSIDER

$$\mathrm{TRIANG}(t) \rightleftharpoons \mathrm{SINC}^2(f)$$

DUALITY PROPERTY: $\mathrm{SINC}^2(t) \rightleftharpoons \mathrm{TRIANG}(-f) = \mathrm{TRIANG}(f)$

($\because$ TRIANG IS A SYMMETRIC FUNCTION)

$$\tau\, \mathrm{sinc}^2(\tau x) \longleftrightarrow \Delta\, \frac{t}{\tau}$$

$\tau = 2$

$$2\, \mathrm{sinc}^2(2t) \longleftrightarrow \Delta\left(\frac{t}{2}\right)$$

SCALING PROPERTY: $\underbrace{\mathrm{SINC}^2(2t)}_{g(t)} \rightleftharpoons \underbrace{\frac{1}{2}\, \mathrm{TRIANG}(f/2)}_{G(f)}$

COROLLARY OF THE FREQ SHIFTING PROPERTY:

$$g(t)\sin(2\pi f_c t) \rightleftharpoons \frac{1}{2j}\left[G(f-f_c) - G(f+f_c)\right]$$

LET $g(t) = \mathrm{SINC}^2(2t)$ & $f_c = 3$ Hz.

$$\mathrm{sinc}^2(2t)\sin(6\pi t) \rightleftharpoons \frac{1}{2j}\left[\frac{1}{2}\,\mathrm{TRIANG}\left(\frac{f-3}{2}\right) - \frac{1}{2}\,\mathrm{TRIANG}\left(\frac{f+3}{2}\right)\right]$$

LET $y(t) = 4\,\mathrm{sinc}^2(2t)\sin(6\pi t) \rightleftharpoons \frac{1}{j}\left[\mathrm{TRIANG}\left(\frac{f-3}{2}\right) - \mathrm{TRIANG}\left(\frac{f+3}{2}\right)\right]$

$= Y(f)$

TIME SHIFTING PROPERTY:

IF $y(t) \rightleftharpoons Y(f)$

$$y(t-t_0) \rightleftharpoons Y(f)\cdot e^{-j2\pi f t_0}$$

LET $t_0 = 5$ s.

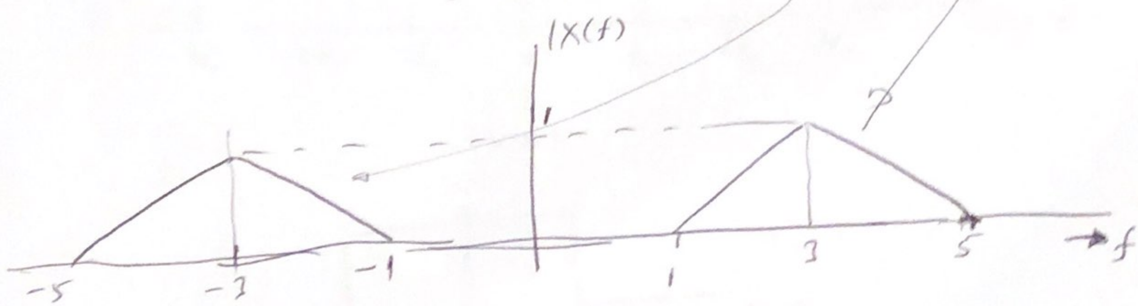
$$x(t) = y(t-5) = 4\,\mathrm{sinc}^2[2(t-5)]\,\sin[6\pi(t-5)]$$

$$x(t) = y(t-5) \rightleftharpoons Y(f)\cdot e^{-j2\pi\cdot f\cdot 5} = X(f)$$

$$\therefore X(f) = \frac{1}{j}\left[\mathrm{TRIANG}\left(\frac{f-3}{2}\right) - \mathrm{TRIANG}\left(\frac{f+3}{2}\right)\right]e^{-j2\pi f\cdot 5}$$

$$|X(f)| = \underbrace{\frac{1}{|j|}}_{=1}\left|\mathrm{TRIANG}\left(\frac{f-3}{2}\right) + \mathrm{TRIANG}\left(\frac{f+3}{2}\right)\right|\underbrace{\left|e^{-j2\pi\cdot f\cdot 5}\right|}_{=1}$$

$$= \mathrm{TRIANG}\left(\frac{f-3}{2}\right) + \mathrm{TRIANG}\left(\frac{f+3}{2}\right)$$



~~PHASE SPECTRUM~~

$\cancel{X(f) = |X(f)|\,e^{j\theta(f)}}$

2.

$T_0 = 0.001s$ = PERIOD

$f_0 = \frac{1}{T_0} = 1kHz$ = FUNDAMENTAL FREQ.

FOR 50% DUTY CYCLE SQUARE WAVE:

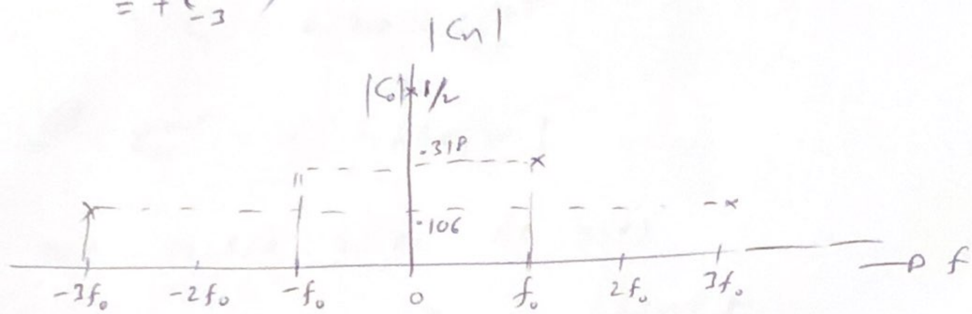
(a) $C_n = \frac{1}{2} \text{sinc}(n/2)$ (REFER TO NOTES)

$C_0 = \frac{1}{2} \text{sinc}(0) = 1/2$

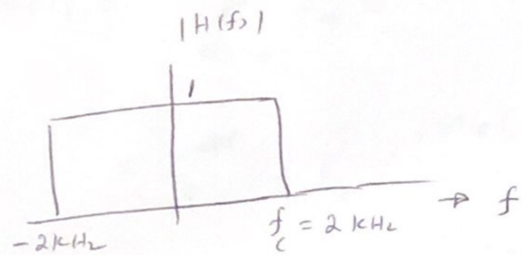
$C_1 = 0.318 = C_{-1}$

$C_2 = 0$

$C_3 = -0.106$
$= + C_{-3}^*$



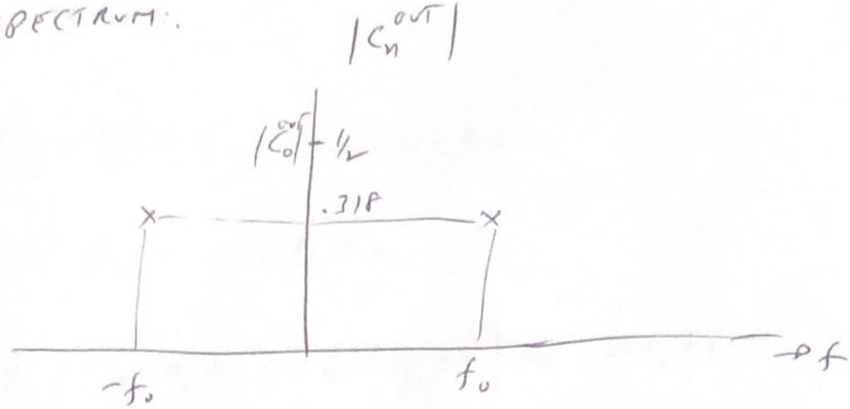
(b)



THE HIGHEST FREQ. COMPONENT THE LPF WILL ALLOW TO PASS = 2 KHz..

$\therefore$ ONLY FUNDAMENTAL FREQ. COMPONENT ($f_0 = 1kHz$) WILL PASS THROUGH THE FILTER UNATTENUATED & THE THIRD HARMONICS & BEYOND ARE FULLY ATTENUATED.

OUTPUT SPECTRUM:



(c)

$$C_0^{OUT} = \frac{1}{2}$$

$$C_1^{OUT} = C_1 = 0.318 = C_{-1}^{OUT}$$

$$C_n^{OUT} = 0 \quad \text{if } |n| > 1$$

LET THE FILTER OUTPUT BE $y(t)$

$$y(t) = \sum_{n=-\infty}^{\infty} C_n^{OUT} e^{j2\pi n f_0 t}$$

$$= C_0^{OUT} + C_1 e^{j2\pi f_0 t} + C_{-1} e^{-j2\pi f_0 t}$$

$$= \frac{1}{2} + 0.318 \left[e^{j2\pi f_0 t} + e^{-j2\pi f_0 t} \right]$$

$$= \frac{1}{2} + 0.318 \times 2 \cdot \cos(2\pi \times 1 \times 10^2 t)$$

$$= 0.5 + 0.636 \cos(2\pi \times 10^2 t)$$

3.

AM WAVE

$$s(t) = \left[1 + k_a m(t)\right] A_c \cos(2\pi f_c t)$$

$$m(t) = A_m \sin(2\pi f_m t)$$

$$s(t) = A_c\left[1 + \underbrace{k_a A_m}_{M} \sin(2\pi f_m t)\right] \cos(2\pi f_c t)$$

$$= \underbrace{A_c \cos(2\pi f_c t)}_{\text{CARRIER}} + A_c M \overbrace{\sin(2\pi f_m t)}^{A} \overbrace{\cos(2\pi f_c t)}^{B}$$

$$s(t) = \underbrace{A_c \cos(2\pi f_c t)}_{\text{CARRIER}} + \frac{A_c M}{2} \Big\{ \underbrace{\overbrace{\sin(2\pi(f_c + f_m)t)}^{A+B}}_{\text{USB}} + \underbrace{\overbrace{\sin\left[2\pi(f_m - f_c)t\right]}^{A-B}}_{\text{LSB}} \Big\} \rightarrow (*)$$

$$\text{CARRIER POWER} = A_c^2/2$$

$$\text{LSB POWER} = (A_c^2 M^2/4)/2 = A_c^2 M^2/8$$

$$\text{RATIO} = \frac{A_c^2 M^2/8}{A_c^2/2} = M^2/4$$

$$k_a\, m(t) = k_a A_m \sin(2\pi f_m t)$$

$$\max\left[k_a m(t)\right] = k_a A_m = M.$$

$$\%\ \text{MODULATION} = \max\left[k_a m(t)\right] \times 100 = 50$$

$$\Rightarrow \quad M = 1/2$$

$$\text{RATIO} = \frac{1}{16}$$

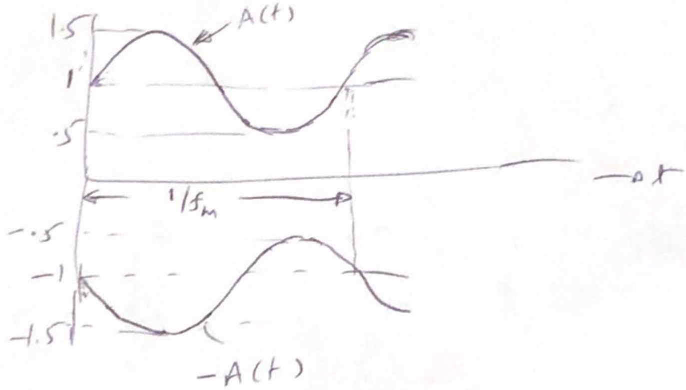
TIME DOMAIN:

$$s(t) = A(t) \cos(2\pi f_c t)$$

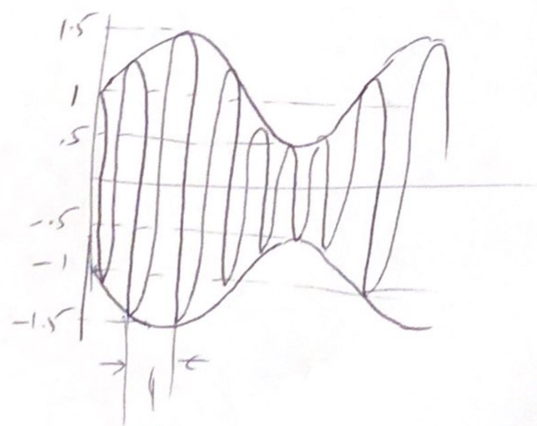
$$A(t) = 1 + M \sin(2\pi f_m t)$$

$$= 1 + 0.5 \sin(2\pi f_m t)$$

STEP 1: SKETCH THE ~~ENVELOPE~~ ENVELOPES $A(t)$ & $-A(t)$



STEP 2: SHOW THE FAST OSCILLATIONS DUE TO CARRIER



FREQUENCY DOMAIN:

Let $s(t) = s_1(t) + s_2(t) + s_3(t)$

From (*), $s_1(t) = A_c \cos(2\pi f_c t)$

$$S_1(f) = \frac{A_c}{2}[\delta(f-f_c) + \delta(f+f_c)]$$

USB: $s_2(t) = \frac{A_c M}{2} \sin[2\pi\underbrace{(f_c + f_m)}^{f^+}t]$

$$S_2(f) = \frac{A_c M}{2}\left\{ \frac{1}{2j}\left[\delta(f-f^+) - \delta(f+f^+)\right]\right.$$

$$f^+ = f_c + f_m$$

LSB: $s_3(t) = \frac{A_c M}{2} \sin[2\pi\overbrace{(f_c - f_m)}^{f^-}t]$

$$S_3(f) = \frac{A_c M}{2}\left\{ \frac{1}{2j}\left[\delta(f-f^-) - \delta(f+f^-)\right]\right\}$$

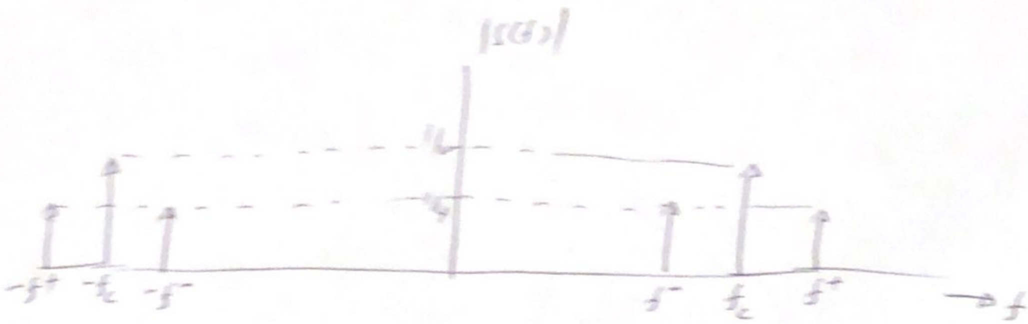
$f^- = f_c - f_m$

MAGNITUDE SPECTRUM:

$A_c = 1V.$

$$|S(f)| = |S_1(f)| + |S_2(f)| + |S_3(f)|$$

$$= \frac{1}{2}[\delta(f-f_c) + \delta(f+f_c)]$$
$$+ \frac{1}{4}[\delta(f-f^+) + \delta(f+f^+)]$$
$$+ \frac{1}{4}[\delta(f-f^-) + \delta(f+f^-)]$$



|S(f)|
$-f^+$
$-f_c$
$-f^-$
f^-
f_c
f^+
f